

$$Q^*(A, \text{stay}) = 1 + 0.8 V^*(A)$$

- Move sends you to B :

$$Q^*(A, \text{move}) = 0 + 0.8 V^*(B)$$

From state B

- Stay keeps you in B :

$$Q^*(B, \text{stay}) = 1 + 0.8 V^*(B)$$

- Move sends you to A :

$$Q^*(B, \text{move}) = 0 + 0.8 V^*(A)$$

And

$$V^*(A) = \max\{Q^*(A, \text{stay}), Q^*(A, \text{move})\}$$

$$V^*(B) = \max\{Q^*(B, \text{stay}), Q^*(B, \text{move})\}$$

3. Use symmetry / solve for V^*

The MDP is symmetric in A and B , so in the optimal policy we'll have

$$V^*(A) = V^*(B) = V.$$

Then the four equations become:

$$Q^*(s, \text{stay}) = 1 + 0.8V, \quad Q^*(s, \text{move}) = 0.8V$$

for $s \in \{A, B\}$.

Now

$$V = \max\{1 + 0.8V, 0.8V\} = 1 + 0.8V$$

because $1 + 0.8V > 0.8V$.

Solve:

$$V = 1 + 0.8V \Rightarrow V - 0.8V = 1 \Rightarrow 0.2V = 1 \Rightarrow V = \frac{1}{0.2} = 5.$$